WG 3

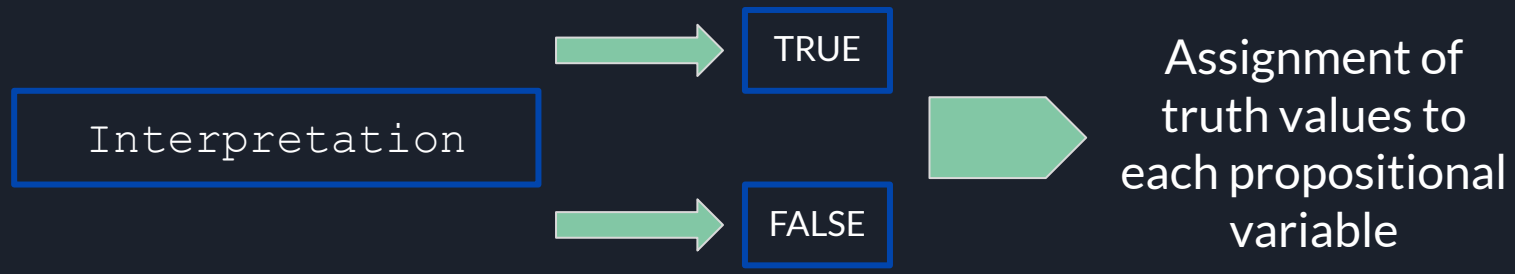
Intelligent Systems

Exercise 1: Consider the following propositional formula: $\neg p \rightarrow (q \,\&\, \neg r)$. Which of the following interpretations is a model?

1. p=True, q=True, r=True
2. p=False, q=False, r=True
3. p= False, q=True, r=True
4. p= False, q=False, r=False

DEFINITION
What is a **model**?

An interpretation of the variables in a formula that makes the formula true.



RULES RECAP

Φ	$\neg\Phi$
T	F
F	T

Φ	Ψ	$(\Phi \rightarrow \Psi)$
T	T	T
T	F	F
F	T	T
F	F	T

Φ	Ψ	$(\Phi \wedge \Psi)$
T	T	T
T	F	F
F	T	F
F	F	F

$$\neg p \rightarrow (q \ \& \ \neg r)$$

Interpretation 1: p=True, q=True, r=True

a. ¬True → (True & ¬True)

b. False → (True & False)

c. <u>False → False = TRUE</u>

$$\neg p \rightarrow (q \ \& \ \neg r)$$

Interpretation 2: p=False, q=False, r=True

a. ¬False → (False & ¬True)

b. True → (False & False)

c. True → False = FALSE

$$\neg p \rightarrow (q \ \& \ \neg r)$$

Interpretation 3: p=False, q=True, r=True

a. ¬False → (True & ¬True)

b. True → (True & False)

c. True → False = FALSE

$$\neg p \rightarrow (q \ \& \ \neg r)$$

Interpretation 4: p=False, q=False, r=False

a. ¬False → (False & ¬False)

b. True → (False & True)

c. True → False = FALSE

Exercise 2:

Write the truth table for formula *A v (B & -A)* by filling in the gaps:

A	B	B & ¬A	A v (B & ¬A)
True	True		
True	False		
False	True		
False	False		

DEFINITION

TRUTH TABLE

A table used in logic to display all possible truth values of its variables.

RULES RECAP

Φ	$\neg\Phi$
T	F
F	T

Φ	Ψ	$(\Phi \rightarrow \Psi)$
T	T	T
T	F	F
F	T	T
F	F	T

Φ	Ψ	$(\Phi \wedge \Psi)$
T	T	T
T	F	F
F	T	F
F	F	F

Φ	Ψ	$(\Phi \vee \Psi)$
T	T	T
T	F	T
F	T	T
F	F	F

A	B	B & ¬A	A v (B & ¬A)
True	True	False	True
True	False	False	True
False	True	True	True
False	False	False	False

Exercise 3: Write the truth table for formula $(A \rightarrow (B \rightarrow A)) \rightarrow (\neg A \vee \neg B)$. Is the sentence valid, (only) satisfiable, unsatisfiable, or none of those?

DEFINITIONS

VALID

True for all interpretations.

SATISFIABLE

True for at least one interpretation.

UNSATISFIABLE

No interpretation makes the formula true.

A	B	(B -> A)	(A -> (B -> A))	(-A v -B)	(A ->(B -> A))->(-A v -B)
True	True	True	True	False	False
True	False	True	True	True	True
False	True	False	True	True	True
False	False	True	True	True	True

SATISFIABLE!

Exercise 4: In Propositional Logic, which of the following formulas is entailed by the knowledge base containing two formulas **p v (¬q & r)** and **(¬r v q)?**

1. p
2. q
3. r
4. none of the above

DEFINITION

KNOWLEDGE BASE

Set of propositional logic formulas (known information).

LOGICAL ENTAILMENT ⊨

If the KB is true, then the formula must also be true.

When the KB (last column) is true, also p is true.

Then, **p is entailed by the knowledge base.**

p	q	r	p v (¬q & r)	(¬r v q)	p v (¬q & r) and (¬r v q)
T	T	T	T	T	T
T	T	F	T	T	T
T	F	T	T	F	F
T	F	F	T	T	T
F	T	T	F	T	F
F	T	F	F	T	F
F	F	T	T	F	F
F	F	F	F	T	F

Exercise 5: In Propositional Logics check whether the following entailment holds:

$$p \vee (\neg q \,\&\, r) \vDash (q \vee \neg r) \rightarrow p$$

1. yes
2. no
3. Cannot be decided
4. It's a tautology

TRY BY YOURSELF!

HINT: Build the truth table and check whether all the models for p v (¬q & r) are also models for (q v ¬r) -> p

SPOILER: Yes, the formula is entailed.

Exercise 6:

Rewrite the following logical formulas in their corresponding Clause Normal Form (CNF)

1. $p \rightarrow q$
2. $\neg (p \vee q)$
3. $\neg p \vee q$
4. $\neg (p \& q)$
5. $\neg p \& q$
6. $r \vee (p \& q)$
7. $A \vee \neg (B \& C)$
8. $(A \vee B) \& (A \vee C)$
9. $A \& (\neg B \vee C)$
10. $A \rightarrow (\neg B \& C)$
11. $\neg(A \vee B) \& C$

DEFINITIONS

CLAUSE

Disjunction (OR) of literals. Basic units of logic that can be combined through conjunctions (AND) to form larger logical expressions.

LITERAL

Either a propositional variable or its negation.

CLAUSE NORMAL FORM (CNF)

Way of structuring formulas such that they are a conjunction (AND) of clauses.

RULES

ELIMINATE EQUIVALENCES

$(A \leftrightarrow B)$ is equivalent to $(A \rightarrow B) \wedge (B \rightarrow A)$.

ELIMINATE IMPLICATIONS

$(A \rightarrow B)$ is equivalent to $(\neg A \vee B)$

MOVE NEGATIONS INWARD (De Morgan's Law)

- $\neg(A \wedge B)$ is equivalent to $\neg A \vee \neg B$
- $\neg(A \vee B)$ is equivalent to $\neg A \wedge \neg B$.

DISTRIBUTE OR over AND

Apply distributive law to ensure the formula is a conjunction of disjunctions.

For instance, transform $A \vee (B \wedge C)$ into $(A \vee B) \wedge (A \vee C)$.

SOLUTION

1. $p \rightarrow q$	1. $\neg p \vee q$		
2. $\neg (p \vee q)$	$\neg p \wedge \neg q$		
3. $\neg p \vee q$	3. $\neg p \vee q$		
4. $\neg (p \& q)$	4. $\neg p \vee -q$		
5. $\neg p \& q$	5a. $\neg p$	5b. q	$\{\{\neg p\},\{q\}\}$
6. $r \vee (p \& q)$	6a. $r \vee p$	6b. $r \vee q$	
7. $A \vee \neg (B \& C)$	7a. $A \vee \neg B \vee \neg C$		
8. $(A \vee B) \& (A \vee C)$	8a. $A \vee B$	8b. $A \vee C$	
9. $A \& (\neg B \vee C)$	9a. A	9b. $\neg B \vee C$	
10. $A \rightarrow (\neg B \& C)$	10a. $\neg A \vee \neg B$	10b. $\neg A \vee C$	(via $-A \vee (-B \& C)$)
11. $\neg(A \vee B) \& C$	11a. $\neg A$	11b. $\neg B$	11c. C

Exercise 7: Let KB be the following knowledge base:

$$KB = \{(P \vee Q) \,\&\, (Q \rightarrow R) \,\&\, (R \rightarrow \neg P), \neg (P \Leftrightarrow \neg Q)\}.$$

Prove or disprove whether KB is satisfiable using DPLL.

SOLUTION STEPS:

1) Start from a ground CNF formula
2) Try to build an assignment that varies the formula
3) The assignment is built using a backtracking mechanism
4) If all clauses are satisfied, the formula is satisfiable with the current assignments.
5) If all possible assignments have been tried and a contradiction was reached each time, the formula is unsatisfiable.

STEP 1

$$(P \vee Q) \,\&\, (Q \rightarrow R) \,\&\, (R \rightarrow \neg P)$$

1) $P \vee Q$

2) $\neg Q \vee R$

3) $\neg R \vee \neg P$

$$\neg (P \Leftrightarrow \neg Q)$$

4) $P \vee \neg Q$

5) $Q \vee \neg P$

step 2: ASSIGNMENTS

p = False

1) Q
2) ¬Q V R
3) ¬R V ¬P
4) ¬Q
5) Q V ¬P

Q is a unit-clause, so we decide to set Q=True. Simplifying the clauses gives:

2) R
4) False (officially called {})

Which is a contradiction. So we need to backtrack on **P=True.**

step 3: ASSIGNMENTS/BACKTRACKING

p = True

Remove P V Q and P V ¬Q, since they are always True.

1) ¬Q V R
2) ¬R
3) Q

Q IS A UNIT CLAUSE, SO Q=True.

1) R
2) ¬R

IT LEADS TO A CONTRADICTION. THIS MEANS THE CLAUSES ARE UNSATISFIABLE.

Exercise 8: Prove your answer from exercise 5 using DPLL by refutation.

DEFINITION

DPLL by refutation

KB ⊨ S if and only if KB together with the **negation of S** is **unsatisfiable** (proof by refutation).

TRY BY YOURSELF

No worries, follow the steps below and you should be fine...

1) Transform p v (¬q & r) into CNF
2) Negate (q v ¬r) -> p
3) Bring to CNF as well
4) Apply classic DPLL
5) Is there a way to satisfy all clauses simultaneously? If not, then we conclude that the formula is entailed by the KB.